Program : Diploma in Renewable Energy	
Course Code : **5298**	Course Title: **Power Electronics Lab**
Semester : **5**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L: 0 T: 0 P: 3)**	Periods per semester: **45**

Course Objectives:

- Impart practical knowledge for the design and setup of different power electronic converters and its application for motor control
- Simulate the various power electronics converters,

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basics of operations of different power electronics devices Concepts of Electrical Power and Energy		Fundamentals of Power Electronics Fundamentals of Electrical & Electronics Engineering	

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Experiment with different power electronic devices to obtain their characteristics.	12	Applying
CO2	Apply different triggering methods to turn on an SCR.	9	Applying
CO3	Develop single phase rectifier and AC voltage controller circuits.	12	Applying

CO4	Identify the operation of electric drives using modern simulation tools.	6	Applying
	Lab Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	**Experiment with different power electronic devices to obtain their characteristics.**		
M1.01	Familiarization and testing of various power electronic devices	3	Applying
M1.02	Plot the static characteristics of SCR	3	Applying
M1.03	Develop the static characteristics of MOSFET	3	Applying
M1.04	Plot the static characteristics of IGBT	3	Applying
CO2	**Apply different triggering methods to turn on an SCR.**		
M2.01	Experiment with R firing scheme of an SCR	3	Applying

M2.02	Experiment with RC firing scheme of an SCR	3	Applying
M2.03	Construct UJT firing scheme of an SCR	3	Applying
	Lab Test – I	3	
CO3	**Develop single phase rectifier and AC voltage controller circuits.**		
M3.01	Build an AC voltage controller using TRIAC/DIAC	3	Applying
M3.02	Develop a single phase semi converter with R/RL loads	3	Applying
M3.03	Construct single phase full converter with R/RL loads (Center tapped configuration)	3	Applying
M3.04	Develop single phase full converter with R/RL loads (Bridge configuration)	3	Applying
C04	**Identify the operation of electric drives using modern simulation tools. (Use standard simulation tools like Matlab/Simulink/ Pspice/Lab view / similar softwares)**		
M4.01	Develop a circuit for the speed control of DC motor using chopper	2	Applying
M4.02	Build a speed control circuit of 3-phase induction motor using V/f control method.	1	Applying
M4.03	Model a 1-phase fully-controlled and half-controlled rectifier with R & RL Loads	2	Applying
M4.04	Construct buck/boost/buck-boost converters	1	Applying
	Lab Test – II	3	

Text / Reference

T/R	Book Title/Author
T1	L. Umanand: Power Electronics – Essentials & Applications, Wiley-India

T2	Mohan, Undeland, Robbins: Power Electronics, Converters, Applications & Design, Wiley- India
T3	Muhammad H. Rashid: Power Electronics Circuits, Devices and Applications, Pearson Education

Online Resources

Sl.No	Website Link
1	https://engineeringwrittennotes.blogspot.com/2016/07/static-v-i-characteristics-of-scr.html
2	https://www.coursehero.com/file/126376380/PE-LAB-MANUALdocx/
3	https://www.electronicsmind.com/firing-circuits-of-thyristor-or-scr/
4	https://www.iitk.ac.in/new/power-electronics-laboratory
5	https://www.udemy.com/course/matlab-simulation-power-electronics/

Student Activity

Suggested Open-ended Experiments:

Students can do open ended experiments as a group of 3-5 preferably a simulation

work. There is no duplication in experiments in between groups. This is mainly for the purpose of continuous internal evaluation. Students should prepare a separate report on an open ended experiment of their choice.

Example 1:To simulate 1-phase fully-controlled rectifier fed Separately Excited DC

motor and observe the speed, torque, armature current, armature voltage, source current

waveforms

Example 2:To simulate 1-phase half-controlled rectifier fed Separately Excited DC

motor and observe the speed, torque, armature current, armature voltage, source current

waveforms

Example 3: To simulate 1-phase semi converter fed Separately Excited DC

motor and observe the speed, torque, armature current, armature voltage, source current

waveforms